



STATCAP AI

AI-Enabled Competency Intelligence for India's Official Statistical System

Problem Statement ID | SIH26101

Organization | Ministry of Statistics and Programme

Implementation (MoSPI)

Theme | Blockchain & Cybersecurity

Category | Software

Team Name | CodeOrbit

Team ID | 11221

THE PROBLEM WE SOLVE

Role requirements and learner evidence are not connected to a measurable competency gap before training.

StatCap: gap → iGOT action → validated assessment → updated profile.



CURRENT GAP

Today: content → completion. StatCap: role requirement → validated gap → targeted course → assessment → updated proficiency.

Training need is not tied to measurable competency evidence.

WORKED EXAMPLE

Survey Data Analyst

Required: Sampling = 4/5

Evidence: assessment = 2/5

Gap: 2 points

→ Recommend mapped iGOT course

→ Generate 5 source-grounded MCQs

→ Reassess and update profile

STATCAP CLOSED LOOP

- 1 Role + required competency
- 2 Validated evidence → proficiency
- 3 Gap = required – current, weighted by role priority
- 4 Gap maps to a targeted iGOT course
- 5 Uploaded PDF/manual → RAG → source-grounded MCQs
- 6 Validated assessment updates proficiency → next action

WHAT EXISTS VS WHAT WE ADD

iGOT/LMS: learning content + completion

Generic RAG: document-grounded questions

StatCap adds: role-weighted gap diagnosis + closed-loop profile update.

WHY THE JURY SHOULD CARE

- MoSPI / NSSTA role alignment
- Explainable recommendation reason
- Every MCQ traceable to source text
- Learning evidence changes the next action

TECHNICAL APPROACH

MVP TECHNOLOGY STACK

Frontend

React + TypeScript + Vite • Tailwind • Recharts

API + Auth

Node/Express + TypeScript • JWT/OAuth2 • RBAC

Data

Supabase PostgreSQL • RLS • encrypted storage

AI / OCR

PDF parse + Tesseract → RAG retrieval → Gemini → MCQ JSON

Security

DPDP controls • audit logs • optional credential hash post-MVP

PROOF ARCHITECTURE – ONE EXECUTABLE PATH

1 INPUT

Role map + learner evidence + uploaded PDF/manual

2 GAP ENGINE

required proficiency – validated current proficiency → priority-weighted gap

3A LEARNING

Gap → competency → iGOT course ID
MVP uses local catalog + mocked adapter

3B ASSESSMENT

PDF → retrieve source passage → generate MCQs → cite source → confidence check → validation gate → release
Source passage + confidence + validation gate before release

4 FEEDBACK

Validated score → updated proficiency → next recommended action

FEASIBILITY AND VIABILITY

36-HOUR MVP PLAN

0–6h data model + role map | 6–14h PDF/RAG + MCQs | 14–22h gap engine + iGOT mock | 22–30h UI/auth | 30–36h validation + demo

0–6h

Model

6–14h

RAG

**14–
22h**

Gap

**22–
30h**

UI

**30–
36h**

Demo

VIABILITY

Start with one role + one competency map.
Then add roles after pilot validation and feedback.

Core learning loop remains independent of blockchain.

DEPENDENCIES

iGOT access: conditional
NSSTA competency maps: use one pilot role
Gemini: approved key; fallback = fixed local test prompts
Blockchain: post-MVP credential-integrity layer; not required for core demo

FALLBACKS

- Local iGOT course catalog + mocked API contract
- One fixed role + competency map
- 5-question validation set for MCQs
- Blockchain credential hash shown only as optional post-MVP

RISKS → CONTROL

Integration → mocked adapter
Hallucinated MCQ → source citation + validation gate
Sensitive data → RBAC/RLS + encrypted storage
Over-scope → freeze MVP to one role, one course catalog, one document

IMPACT, SCALE & BENEFITS

PILOT KPIs – MEASURE PRODUCT IMPACT, NOT JUST ECOSYSTEM SIZE

≥30%

faster learning-path discovery

≥20%

competency-gap closure

≥90%

MCQs traceable to source

100%

released MCQs validated

IMPACT

- NSSTA capacity building becomes diagnosis-first: role → gap → targeted learning.
- Assessment evidence updates the competency profile instead of ending at course completion.
- Pilot success is measured by time saved, gap closure and question traceability.

ADDRESSABLE SCALE

iGOT ecosystem context:

17.06M registered Karmayogis • 5,683 courses • 145.32M course completions

These figures show potential reach – they are not claimed StatCap impact.

WHAT EXISTS

- iGOT / LMS
 - content discovery
 - course delivery
 - completion records
- Generic RAG assessment
 - document-grounded questions

WHAT STATCAP ADDS

- role-weighted competency gap
- gap → targeted iGOT action
- source-grounded MCQs
- validated evidence updates profile
- explainable recommendation reason

WHY THIS FITS MOSPI

- competency-based capacity building
- NSSTA-aligned role focus
- privacy/security controls
- optional verifiable credential hash

RELATED WORK / DESIGN BASIS

- [1] Karmayogi Bharat, “iGOT Karmayogi,” Government of India.
- [2] MoHUA, “Karmayogi Guidelines, 2023.”
- [3] MoSPI, “Annual Report 2023–24,” pp. 72–73.
- [4] Ministry of Law & Justice, “Digital Personal Data Protection Act, 2023.”
- [5] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” 2020.
- [6] Polygon documentation / architecture source matching the credential-hash design.